

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Dot .Net Core and Progressive Web Application Practical

Practical – VIII

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Class: TYIT	Batch: 1
Date of Assignment: 06/08/2026	Date/Time of Submission: 06/08/2026

a. Create an HTML page that would be the starting point of the application. This HTML will contain a link to the file named manifest.json. This is an important file that would be created in the next step.

Code:

Student.cs:

```
namespace StudentPWA.Models
{
    public class Student
    {
        public int RollNo { get; set; }
        public string Name { get; set; }
    }
}
```

```
RollNo = 102,
Name = "Asmitha"
});
```

Manifest.json:

```
{
  "name": "Student PWA",
  "short_name": "Student",
  "start_url": "/",
  "display": "standalone"
}
```

Add to _Layout.cshtml:

Inside the <head> section:

```
<link rel="manifest" href="/manifest.json">
```

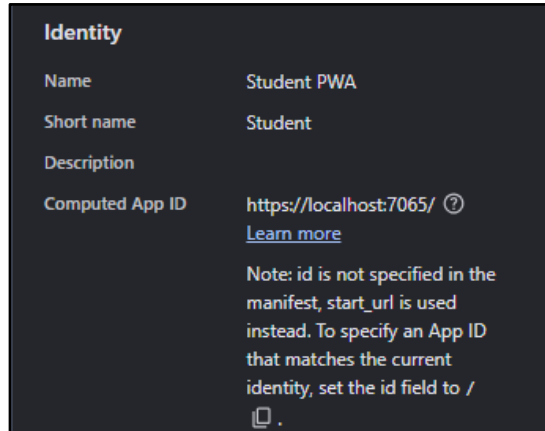
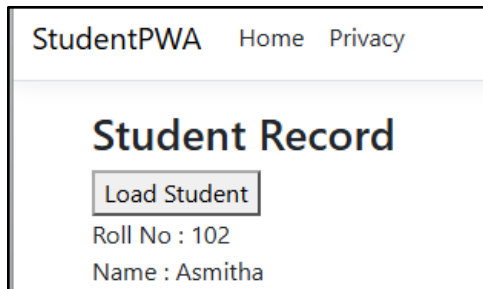
StudentController.cs:

```
using Microsoft.AspNetCore.Mvc;
namespace StudentPWA.Controllers
{
    [ApiController]
    [Route("api / [controller]")]
    public class StudentController : ControllerBase
    {
        [HttpGet]
        public IActionResult Get()
        {
            return Ok(new
            {
            }
        }
    }
}
```

Index.cshtml:

```
<h2>Student Record</h2>
<button onclick="loadStudent()">Load
Student</button>
<p id="data"></p>
<script>
    function loadStudent(){
        fetch("/api/student")
        .then(response => response.json())
        .then(student=>{
            document.getElementById("data").innerHTML=
            "Roll No : "+student.rollNo+
            "<br>Name : "+student.name;
        });
    }
</script>
```

Output:



b. Create a simple Web API that returns Student data in JSON format.

Code:

Student.cs:

```
namespace StudentPWA.Models
{
    public class Student
    {
        public int RollNo { get; set; }
        public string Name { get; set; }
    }
}
```

StudentController.cs:

```
using Microsoft.AspNetCore.Mvc;
namespace StudentPWA.Controllers
{
    [ApiController]
    [Route("api / [controller]")]
    public class StudentController : ControllerBase
    {
        [HttpGet]
        public IActionResult Get()
        {
            return Ok(new
            {
                RollNo = 102,
                Name = "Asmitha"
            });
        }
    }
}
```

```
        RollNo = 102,
        Name = "Asmitha"
    });
}
}
```

Output:

